AAMOD ATRE

M. Sc. Quantum Science and Technology · Technical University of Munich

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Profile

Quantum physicist with hands-on experience in quantum error correction, many-body simulation, and HPC-based benchmarking. Skilled at translating theoretical models into scalable implementations: from stabilizer simulations and semiclassical dynamics to benchmarking HPC-enabled quantum simulators and decoders for production. Focused on applying this combined computational physics and software expertise to scaling practical quantum systems.

Education

M. Sc. Quantum Science and Technology **Oct. 2022 – Jun. 2025**
Technical University of Munich GPA 1.4 / 5.0 | max. 1.0

B. E. Chemical Engineering | Minor: Physics **Aug. 2017 – Jul. 2021**
Birla Institute of Technology and Science, Pilani GPA 8.65 / 10.0 | max. 10.0

Academic Research

Master's Thesis: Measurement-Induced Entanglement Transitions **Apr. 2024 – May 2025**
Prof. Frank Pollmann | Technical University of Munich PDF

Simulated measurement-induced entanglement transitions in random quantum circuits across varying qudit dimensions using the stabilizer formalism. Analyzed transition universality and developed a classical dynamical analogue enabling scalable simulations. Results presented at DPG Spring Meeting 2025.

Undergraduate Thesis: Semiclassical Dynamics of Exciton Relaxation **Jan. 2021 – Jun. 2021**
Prof. Jeremy Richardson | ETH Zürich

Assessed a spin-mapping semiclassical dynamics method via numerical simulation of exciton relaxation in a 1D polymer chain. Developed a spin-mapping algorithm for tight-binding systems with SU(2) symmetry for improved propagation accuracy at consistent resource cost.

Work Experience

Internship | Quantum Error Correction Engineer **Apr. 2026 – May 2026**
QC Design, Ulm

Integrated matching- and belief-propagation-based decoder algorithms into QC Design's hardware-oriented fault-tolerant circuit design software. Estimated decoding thresholds across multiple code families to evaluate performance trade-offs.

Research Assistant (HiWi) | HPC-QC Integration **May 2024 – Sep. 2025**
Dept. of Quantum Computing & Technologies | Leibniz Supercomputing Centre (LRZ), Munich

Benchmarked HPC-enabled quantum simulators on optimisation and Hamiltonian evolution tasks, for integration into LRZ's HPC infrastructure. Investigated automated calibration protocols for the Munich Quantum Software Stack (MQSS), evaluating suitable infrastructure for the calibration pipeline.

Research Assistant (HiWi) | Open Quantum Systems **Apr. 2023 – Sep. 2023**
Prof. Peter Rabl | Walther Meißner Institute, Munich

Compared numerical integration schemes in stochastic master equation simulations of a dissipative cascading quantum network, enabling reliable simulation of larger network sizes.

Remote Research Assistant | Semiclassical Open Quantum Dynamics **Oct. 2021 – Oct. 2022**
Dr. Aaron Kelly | Max Planck Institute for the Structure and Dynamics of Matter, Hamburg

Benchmarked a semiclassical mapping-based dynamics method (spin-PLDM) by modelling the interactions between a two-level atomic subsystem and a cavity-modified field.

Remote Research Intern | Semiclassical Quantum Dynamics **Jul. 2021 – Sep. 2021**
Prof. Pengfei Huo | University of Rochester, New York

Numerically compared standard and spin-based partially linearized density matrix (PLDM) algorithms for computing linear absorption spectra of a bi-exciton coupled dimer model.

Research Intern | Computational Chemistry **May 2019 – Jul. 2019**
Prof. Bibek Dash | CSIR-IMMT, India

Designed triazole-based molecular precursors for CO_2 capture using DFT; identified optimal functional–basis set combinations for modelling CO_2–triazole interactions and proposed improved triazole moiety designs based on interaction analysis.

Technical Skills

Programming	Python, C++, Julia, MATLAB
Quantum Computing	Qiskit (Aer, Experiments), Stim, QuTiP
Scientific Computing	HPC parallel computing (SLURM, MPI), tensor network methods, Monte Carlo methods, stochastic ODE/PDE solvers, *ab initio* molecular modelling (Quantum Espresso, Gaussian)
Languages	English (native/bilingual), German (B1), Hindi & Marathi (native), French (elementary)

Extracurricular Activities

Berlin Quantum Hackathon	Competitively selected; contributed to crew scheduling optimisation using multi-objective QAOA	**Dec. 2025**
QOSF Cohort 4	Implemented multi-objective Grover's search to pass competitive screening assessment	**Sep. 2021**
IBM Quantum School	Completed intensive training in quantum algorithms, noise modelling, and Hamiltonian simulation	**2020 – 2022**
PushQuantum	Organised company visits within Munich's quantum startup space.	**2023 – 2025**

Undergraduate Work

Study Project	Literature survey of cavity QED formalism, Jaynes–Cummings model, and qubit implementations in molecule-coupled cavity systems	**2020**
Design Project	*Ab initio* study of graphene-based Li-MOFs for hydrogen production: DFT-based electronic structure calculations (Quantum Espresso) and reactive molecular dynamics (LAMMPS)	**2020 – 2021**
Teaching	Teaching Assistant: Quantum Mechanics II and Process Design Principles I, BITS Pilani	**2020**
Standardized Tests	GRE 335/340 | Quant. 168/170 | Verbal 167/170 | AW 5/6	**Aug. 2021**